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Coupled Stress Release Model for Time-dependent Seismicity

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Abstract—Based on the original stress release model of seismicity proposed by VERE-JONES (1978), this paper has developed a stochastic coupled stress release model of time-dependent seismicity, which considers the earthquake interaction and stress transfer between different seismic subregions. As an example, the model is applied to a statistical analysis of the historical earthquake catalog with magnitude $M \geq 6.0$ during the period from 1480 to 1996 in North China. According to the Akaike information criterion (AIC), the results show that the coupled stress release model is better than the original model, which demonstrates the existence of long-range correlations between different seismic subregions. We also apply both the stochastic (original and developed coupled) models to analyze the synthetic catalog produced by a cellular automata model, which is based on mechanics of a slide-spring-damper system to model the fault network. The stress release model provides a good fit to the synthetic regional stress, and the coupled stress release model provides an improvement in fit to the synthetic catalog over the original model.

Key words: Coupled stress release model, AIC criterion, historical earthquake catalog, synthetic catalog.

1. Introduction

The stress release model was first proposed by Vere-Jones in 1978 (VERE-JONES, 1978), which is used to study the statistical regularity of seismicity. The physical essence of this model is elastic rebound theory. The classical elastic rebound model of earthquake occurrence considers that the stress in a region is accumulated slowly over a prolonged time, afterward, the stress would be suddenly released and an earthquake would occur when the stress exceeds the strength of the medium. In the field of stochastic process, this conception can be simulated using a leap-type Markov process (KNOPOFF, 1971). The stress release model is a development of Knopoff's Markov model. It uses a notional regional stress level as the key variable, or state, that controls the probabilities of earthquakes within a region.

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VERE-JONES and DENG (1988) applied this model to the historical earthquake catalog in North China and obtained some definite results. ZHENG and VERE-JONES (1991, 1994) studied this model in detail, developed a complete and concrete computation method, and applied this model to the historical earthquake catalogs in China, Japan, Iran, etc. They also conducted further discussion on their results. However, they only considered the condition of one region. Although ZHENG and VERE-JONES (1991) divided North China into four seismic belts and combined them to calculate the parameters of the stress release model, they did not consider the spatial interaction of earthquake occurrence through stress transfer between seismic belts (or different parts) in the model. In this paper, a coupled stress release model is proposed by adding stress interaction between different parts to estimate the effect on the seismic hazard in each part, based on the original stress release model, and the coupled model is applied to analyze the historical earthquake catalog in North China. Based on the Akaike information criterion, a comparison study is done by studying the results from various models.

Due to the time-length limitation of the historical earthquake catalog, we also apply the stochastic models to analyze the synthetic catalog produced by a cellular automata model. A cellular automata model was first proposed by BAK and TANG in 1989, which considered earthquake occurrence as a typical self-organized critical phenomena (BAK and TANG, 1989; ITO and MATSUZAKI, 1990; MAIN, 1996). An essential assumption in their models is that stress increases in each cell purely randomly. This assumption is not appropriate for a real earthquake. Tectonic stress field varies according to mechanical laws, not randomly. A slide-spring-damper mechanical model of a continental earthquake produced by a network of faults was proposed by SHI *et al.* (1994). Based on the mechanical model, the global rules of stress adjustment in the system can be summarized, and correspondingly a similar cellular automata model can be constructed (LIU *et al.*, 1995). Both the fault network model and corresponding cellular automata model cannot only produce chaotic synthetic catalogs, but also simulate the stress evolution of the system. In this paper, we analyze the synthetic earthquake catalogs by applying the stochastic models. The results indicate that even if the model and parameters are unknown, some characteristics of the system, such as the system stress, can be predicted from the synthetic catalog alone. The simplified stress is useful for prediction of the earthquake in the synthetic catalog.

2. *Coupled Stress Release Model*

2.1. *Simple Stress Release Model*

Stress level can be simplified as a scalar function $X(t)$ in the studied region

$$X(t) = X(0) + \rho t - S(t) \tag{1}$$

where ρ is a constant loading rate from external tectonic force. $S(t)$ is the accumulated stress release from events within the region over the period $(0, t)$

$$S(t) = \sum_{0 < t_i < t} S_i \tag{2}$$

assume that stress drop is only related to magnitude

$$S_i = 10^{0.75(m_i - m_0)} \tag{3}$$

where m_i is earthquake magnitude. m_0 is considered as normalized magnitude, which is a constant.

The stochastic behavior of the model is controlled by a "risk function" $\Psi(x)$. It represents the conditional probability intensity on condition that $\Psi(x)$ is equal to x (x is a stress value), or $\Psi(x)\,dt$ is the occurrence rate in the expected time interval $(t, t + dt)$. In the Poisson model, seismic risk is always considered as a constant regardless the stress value, but, in the stress release model, the seismic risk function is assumed to exponentially increase with stress, i.e.,

$$\Psi(x) = A \exp(d + bx). \tag{4}$$

In the above assumption, the key element of the stochastic point process analysis is to find conditional probability intensity function λ, which is also the basis of statistical inference, simulation and prediction

$$\begin{aligned} \lambda(t, s)\,dt\,ds = \text{Prob}\{&\text{event in time interval } (t, t + dt) \\ &\text{and stress drop within } (s, s + ds) \\ &\text{given the past history up to time } t\}. \end{aligned} \tag{5}$$

How to get λ? There is a set of strict theories and methods in statistics. ZHENG and VERE-JONES (1991, 1994) have presented detailed discussion in their papers. The conclusion is that the effect for λ by time and stress can be separated to calculate. The change of λ with time can be written:

$$\lambda(t) = \Psi(X(t)) \tag{6}$$

substitute Equations (1), (2) and (3) into Equations (4) and (5), and obtain Equation (7)

$$\lambda(t) = \exp\{a + b[t - cS(t)]\} \tag{7}$$

where a, b and c are parameters, they can be obtained by achieving $\log L$ maximum using a likelihood method

$$\log L = \sum_{i=1}^{N(T)} \log \lambda(t_i) - \int_0^T \lambda(u)\,du \tag{8}$$

where T is the length of the observation interval (conventionally presumed to start time $t = 0$) and $N(T)$ is the total number of events observed in that interval, t_i is the origin time of each earthquake.

2.2. AIC Criterion

There are many statistical models to be selected for describing a natural phenomenon. If just comparing their log-likelihood values, the conclusion will be that the more parameters, the better the results. However, that is not correct in a practical sense. In order to solve this problem, AKAIKE (1977) proposed the AIC (Akaike Information Criterion) method to select models. The method is: let $\log \hat{L}$ denote the log-likelihood for a given model, so

$$\mathrm{AIC} = -2 \log \hat{L} + 2k \tag{9}$$

where k is the number of fitted parameters in the given regional model. The AIC value can be obtained for different models. The model with the smallest AIC value is considered as the best model to be selected. Generally, if model difference reaches a 5% credible level, these two models can be selected, which corresponds to a difference in AIC values of around $1.5 \sim 2.0$.

2.3. Coupled Stress Release Model

From the above description, it can be seen that a simple stress release model only considers the stress-level effect of a region by the seismicity in the same region, yet, in practical seismicity, the seismicity around this region can also produce a definite effect on stress level in the region. Conversely, its seismicity can cause a definite effect on seismic hazard around this region too. Therefore, a coupled stress release model is proposed in this paper. For simplicity, we only consider coupled interaction in two subregions. Assuming that $\lambda_1(t)$ and $\lambda_2(t)$ are conditional probability intensity functions in the first subregion and the second subregion respectively, they can be described as

$$\begin{aligned} \lambda_1(t) &= \exp\{a_1 + b_1[t - c_{11}S_1(t) - c_{12}S_2(t)]\} \\ \lambda_2(t) &= \exp\{a_2 + b_2[t - c_{21}S_1(t) - c_{22}S_2(t)]\} \end{aligned} \tag{10}$$

where $a_1, b_1, c_{11}, c_{12}, a_2, b_2, c_{21}, c_{22}$ are model parameters. The total number is 8. $S_1(t)$ and $S_2(t)$ are the accumulated stress release in the first and second subregions separately. The computation method is the same as Equation (2). Comparing Equation (7) with Equation (10), coupled interaction is realized by introducing the stress release effect of the other subregion by parameters c_{12} and c_{21}. This definition is also simplified as that of stress function. The model parameters can be obtained by using the log-likelihood method

$$\log L = \sum_{i=1}^{N_1(T)} \log \lambda_1(t_i) + \sum_{j=1}^{N_2(T)} \log \lambda_2(t_j) - \int_0^T (\lambda_1(u) + \lambda_2(u))\, du \tag{11}$$

where $N_1(t)$ and $N_2(t)$ are the total earthquake number in the first and second subregions in the time interval T respectively; t_i and t_j are occurrence times of earthquakes in those two subregions separately.

3. Computation Models and Results of Historical Earthquakes

When the statistical model is applied to the historical earthquake catalog in North China, we select only $M_S \geq 6.0$ earthquakes from 1480 to 1996, considering the completeness of the earthquake catalog and referring to the division method for the North China region in Zheng's papers (ZHENG *et al.*, 1991). The subregion is divided, and the boundary line is the Taihangshan Mountains (Fig. 1). The earthquakes in the western subregion are mainly events around the Ordos Plateau (not including earthquakes in the Qilianshan seismic belt and southeast of Gansu Province, and the boundary line is delimited along the edge of North China tectonic plate). The earthquakes in the eastern subregion mainly include the events in the Hebei plain seismic belt and the Tanlu seismic belt. Because each earthquake is considered as an independent event in stochastic point process, aftershocks are deleted from the catalog using space-time window criteria suggested by KEILIS-BOROK and KNOPOFF (1980). Although a satisfactory definition of such a scheme is very difficult, aftershocks are not present in large numbers due to the magnitude cut-off of $M \geq 6.0$ fortunately. After the removal of aftershocks, there remains a total of 65 earthquakes in the entire region, of which there are 33 earthquakes in the eastern subregion and 32 in the western subregion. The resulting data set is provided in the Appendix. The following three models are used in computation.

3.1. Simple Stress Release Model

Simple stress release model is used in the entire North China region, and the random allocation method is used to distribute the earthquakes in the region. Their computation formulas are as follows:

$$\begin{gathered} \lambda(a, b, c) = \exp\{a + b[t - cS(t)]\} \\ \log L_1 = \sum_{i=1}^{N} \lambda(t_i) - \int_0^T \lambda(u)\, du \\ \log L_2 = N_1 \log \frac{N_1}{N_1 + N_2} + N_2 \log \frac{N_2}{N_1 + N_2} \\ \mathrm{AIC} = -2(\log L_1 + \log L_2) + 2 * (3 + 1) \end{gathered} \tag{12}$$

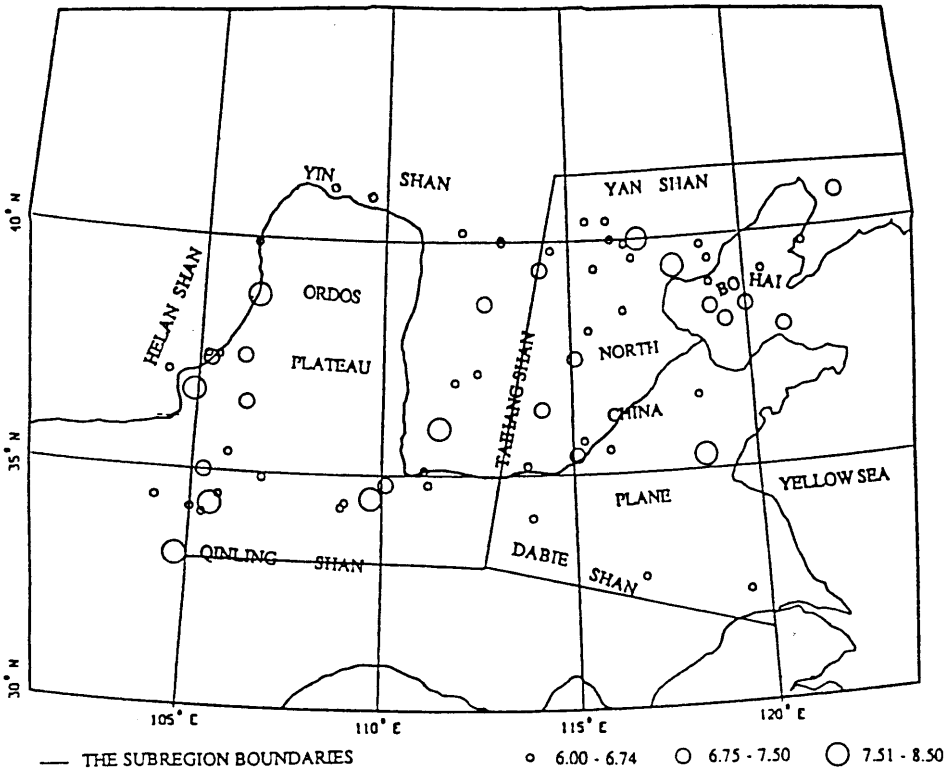

Figure 1
Map of North China showing historical records and subregion boundaries.

where N, N_1 and N_2 is the total earthquake number, the earthquake number in the first and second subregion, respectively. There are three parameters a, b and c in stress release model. Because the total number of earthquakes in the entire region is a fixed number, it needs only one parameter to allocate earthquakes randomly to two subregions. Consequently, the total parameters are 4.

Table 1
Simple stress release model

a	B	c	$-\log L_1$	$-\log L_2$	AIC
−2.462	0.01128	0.1513	195.87	45.05	489.83

Table 2

Independent stress release model

a_1	b_1	c_1	$-\log L_1$	a_2	b_2	c_2	$-\log L_2$	AIC
−3.46	0.0132	0.329	118.13	−3.15	0.0155	0.284	117.86	483.97

3.2. Independent Stress Release Model

In each subregion, simple stress release model is computed independently to obtain the conditional probability intensity function. The conditional probability intensity in the entire region can be gained by simple addition.

$$\begin{gathered}\lambda_1(a_1, b_1, c_1) = \exp\{a_1 + b_1[t - c_1 S_1(t)]\} \\ \log L_1 = \sum_{i=1}^{N_1} \log \lambda_1(t_i) - \int_0^T \lambda_1(u)\, du \\ \lambda_2(a_2, b_2, c_2) = \exp\{a_2 + b_2[t - c_2 S_2(t)]\} \\ \log L_2 = \sum_{j=1}^{N_2} \log \lambda_2(t_j) - \int_0^T \lambda_2(u)\, du \\ \lambda(t) = \lambda_1(t) + \lambda_2(t) \\ \mathrm{AIC} = -2 * (\log L_1 + \log L_2) + 2 * (3 + 3).\end{gathered} \tag{13}$$

There are three parameters in each subregion, thus the total number of parameters is 6.

3.3. Coupled Stress Release Model (CSRM)

For coupled stress release model, its computation method is shown in Equations (10) and (11), considering the interaction between two parts. In addition, we also use a simplified model in this paper which is called a symmetrical coupled stress release model, i.e., assume that interaction within the two subregions is the same, consequently, it is in the equation of conditional probability intensity

$$c_{12} = c_{21}. \tag{14}$$

Thus there are only seven parameters in symmetrical CSRM. The equations to calculate AIC values of the two kinds of CSRM are separately as follows:

$$\begin{gathered}\mathrm{AIC} = -2 * \log L + 2 * (4 + 4) \text{ (for common CSRM)} \\ \mathrm{AIC} = -2 * \log L + 2 * 7 \text{ (for symmetrical CSRM).}\end{gathered} \tag{15}$$

In all above computations, normalized magnitude $m_0 = 5.0$. The results are given in Tables 1–3 and Figures 2–4.

When comparing obtained AIC values in the three models, the difference between them is larger than 2. This demonstrates that the difference between models is obvious. Coupled stress release model is better than independent stress release model, or better than the stress release model in which the stress is calculated in each subregion independently and the interaction is not considered. Furthermore, independent stress release model is better than the simple stress release model, in which the whole region is set up as only one stress parameter. This indicates that the result, considering region division is better than the results in the entire region when applying stress release model. Zheng considered region division in his paper (ZHENG *et al.*, 1991), and he remarked mainly on the comparison of various models (Poisson model, Poisson model with trend and simple stress release model) after region division, however he did not reveal how to compare their AIC values between the subregions and the entire region (for example, we introduce the random-distribution parameter in simple stress release model).

When comparing symmetrical CSRM and common CSRM, and AIC value in symmetrical CSRM is smaller than that of common CSRM although the minus log-likelihood value of symmetrical CSRM is larger than that of common CSRM. Because the difference of AIC values is smaller (less than 2), we cannot determine which model is better by the AIC criterion method despite the fact that there is one more parameter in common CSRM than in symmetrical CSRM.

In order to study coupled stress release model further, the North China region is divided into four subregions. In the eastern part, the Hebei plain seismic belt is a subregion (E1), and the other is the Tanlu seismic belt (E2). The total numbers are 21 and 12, respectively. In the western part, the Shanxi seismic belt is the subregion (W1), and the other is the western boundary area of Ordos plateau (W2). The total numbers are 20 and 12 separately (see Appendix). The above three kinds of models are applied to both these two parts. Comparing their difference of AIC values, the same conclusion can be reached (Table 4). In addition, because the tectonic environment is similar in the eastern and western parts of North

Table 3

Coupled stress release model

	a_1, a_2	b_1, b_2	c_{11}, c_{21}	c_{12}, c_{22}	$-\log L$	AIC
Symmetrical	−3.949	0.02578	0.1580	0.1390	232.603	479.21
CSRM	−3.293	0.01680	0.1390	0.1686		
Common	−3.939	0.02632	0.1448	0.1506	232.464	480.93
CSRM	−3.322	0.01804	0.1030	0.1971		

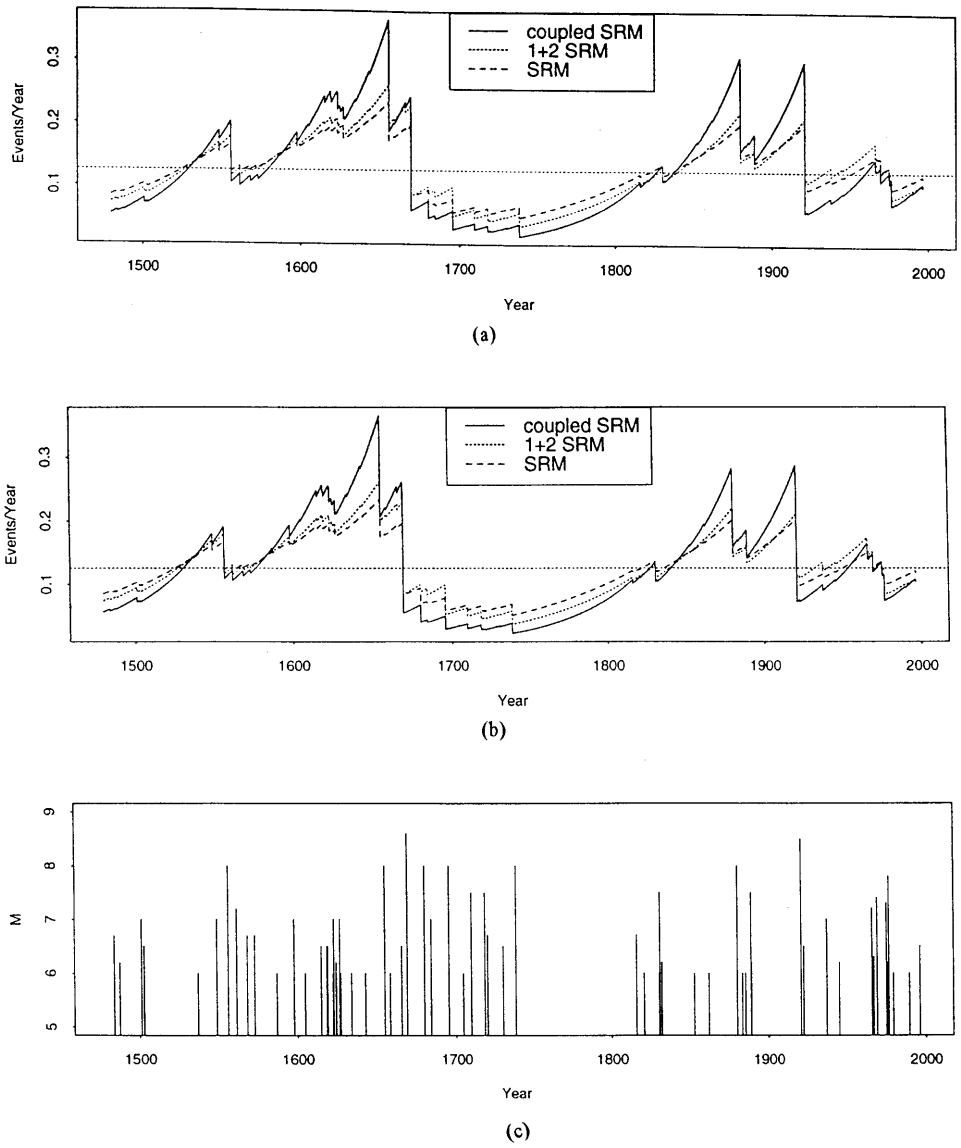

Figure 2
The conditional probability intensity plot of stress release model and M-T plot in North China ($1+2$ SRM means independent stress release model). (a) Coupled stress release model, (b) symmetrical coupled stress release model, (c) magnitude of event versus time (M-t plot).

China (ZHENG *et al.*, 1991), the stress-accumulated condition may be assumed as the same, therefore, another computation method could be proposed for CSRM, i.e.,

$$b_1 = b_2$$
$$\mathrm{AIC} = -2 * \log L + 2 * 7. \tag{16}$$

Comparing the AIC values of these three CSRM models, the difference between common CSRM and symmetrical CSRM is still smaller than 2, yet the difference between the results of both models and the result of $b_1 = b_2$ CSRM is larger than 2. This illustrates that the model assuming there is the same stress-accumulated rate, is not the best model. On the other hand, the result of $b_1 = b_2$ CSRM is smaller than that of independent stress release mode, which establishes that coupled interaction is an important factor in stress release models. However, because there is one subregion in which the earthquake number is small (only 12) for both regions, its credible level is slightly lower from the point of statistics.

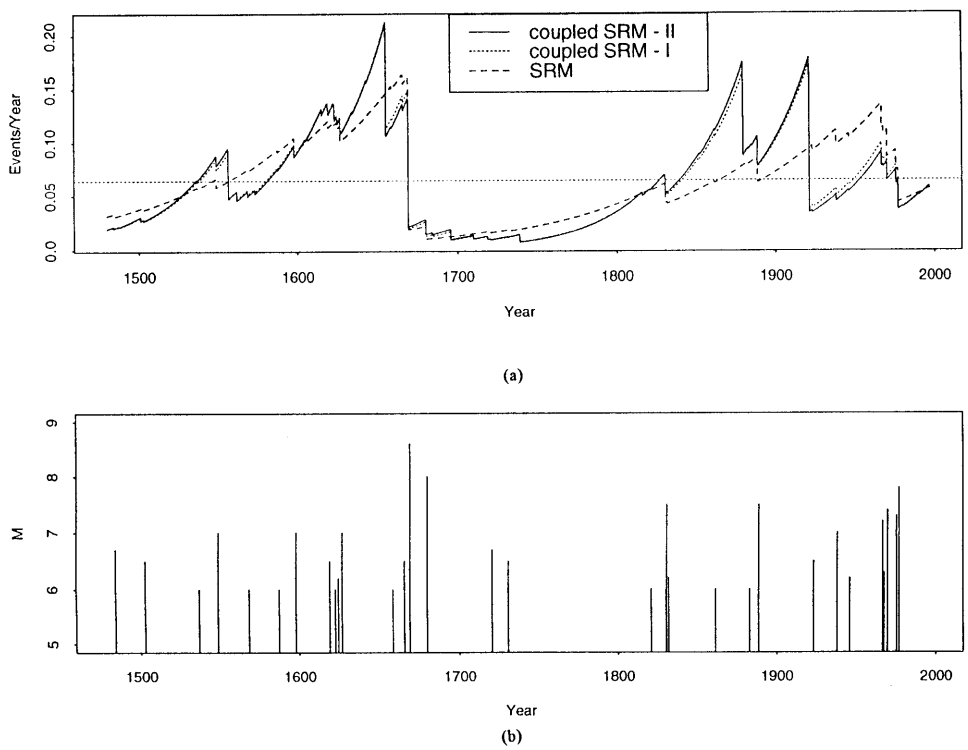

Figure 3
The conditional probability intensity plot of stress release model (a) and M-T plot (b) in the eastern part (the first subregion) of North China (coupled SRM-I is common coupled stress release model, coupled SRM-II is symmetrical coupled stress release model).

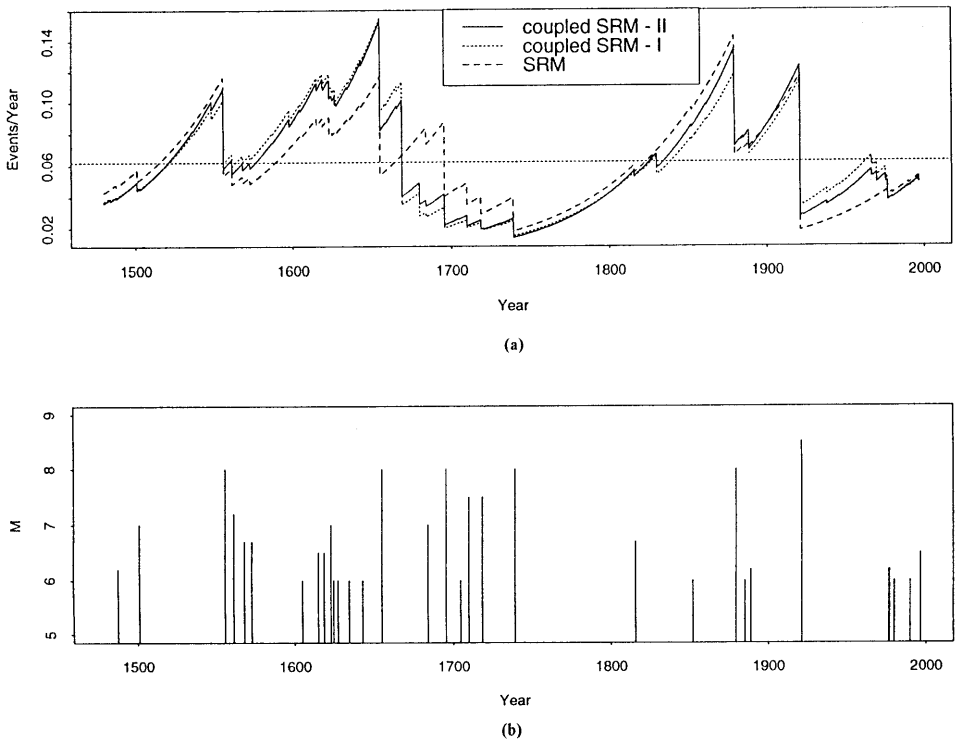

Figure 4
The conditional probability intensity plot of stress release model (a) and M-T plot (b) in the western part (the second subregion) of North China (coupled SRM-I is common coupled stress release model, coupled SRM-II is symmetrical coupled stress release model).

4. *Application of the Stochastic Models to Synthetic Earthquakes*

4.1. *Mechanical Model for Earthquake in a Fault Network*

SHI *et al.* (1994) proposed a spring-slide-damper mechanical model to simulate a fault-network earthquake. By solving a set of differential equations under uniform boundary strain rate conditions, stresses of each element and the entire system can be calculated. An element will break as its stress exceeds its static strength, and the stress will begin to drop until it reduces to its dynamic strength. Potential energy released during the earthquake defines the magnitude of the earthquake. Stress adjustment after an earthquake can be calculated according to the equilibrium condition of the system. A large earthquake may occur if the adjustment of stress induces further break adjacent to the broken element. A synthetic catalog simulates many aspects of continental earthquakes produced by such computation (ZHANG *et al.*, 1995).

Based on the rules of stress adjustment summarized from the network model, LIU *et al.* (1995) proposed a corresponding cellular automata model. The model consists of an array of element. A column of elements represents a fault, and different columns represent a group of nearly parallel faults mechanically coupled. In the model under boundary conditions of constant deformation rate, stresses in all elements as well as the stress of the entire system can be calculated. They all increase at a different rate linearly with time until break of an element occurs. Break of an element produces significant stress drop in the element. Nearby elements in the same fault undergo stress increase because of the stress concentration produced by the broken element. The stress increase decays with the distance to the broken element. As concerns the entire fault, the summation of stresses in all elements of the fault is still reduced relative to that before the event. Stresses on the other faults sustain the same stress drop, and the amount of stress drop for each element is also related to its distance to the broken element. Continuous breaks induced by a broken element represent a large earthquake.

4.2. Computational Results

A cellular automata model of 2 faults each has 78 elements is calculated. The model contained 620 independent parameters, such as element statistic friction, dynamic friction, stress increase rate, and initial stress, etc. The synthetic catalog produced by the model is similar to real seismicity in many aspects.

Figure 5 displays the catalog and the stresses calculated from the cellular automata model, and the calculated conditional intensity from 141 events spanning the time period 20,000 to 21,000 time unit in the synthetic catalog. The logarithm of conditional intensity can be compared with the stress. It is noted that the two curves look quite similar. The stress is calculated from a cellular automata model with 620 parameters, while the stress release model uses only the catalog to fit 3

Table 4

The results of stress release model in the eastern and western parts of North China

Division	Stress release model		$-\log L$	AIC
The eastern part	Coupled model	$b_1 = b_2$	126.830	267.660
		Symmetrical	116.196	246.392
		Common	114.955	245.909
	Independent model		136.476	284.952
	Simple model		139.758	287.516
The western part	Coupled model	$b_1 = b_2$	122.741	259.482
		Symmetrical	113.550	241.100
		Common	113.365	242.730
	Independent model		135.792	283.584
	Simple model		139.027	286.054

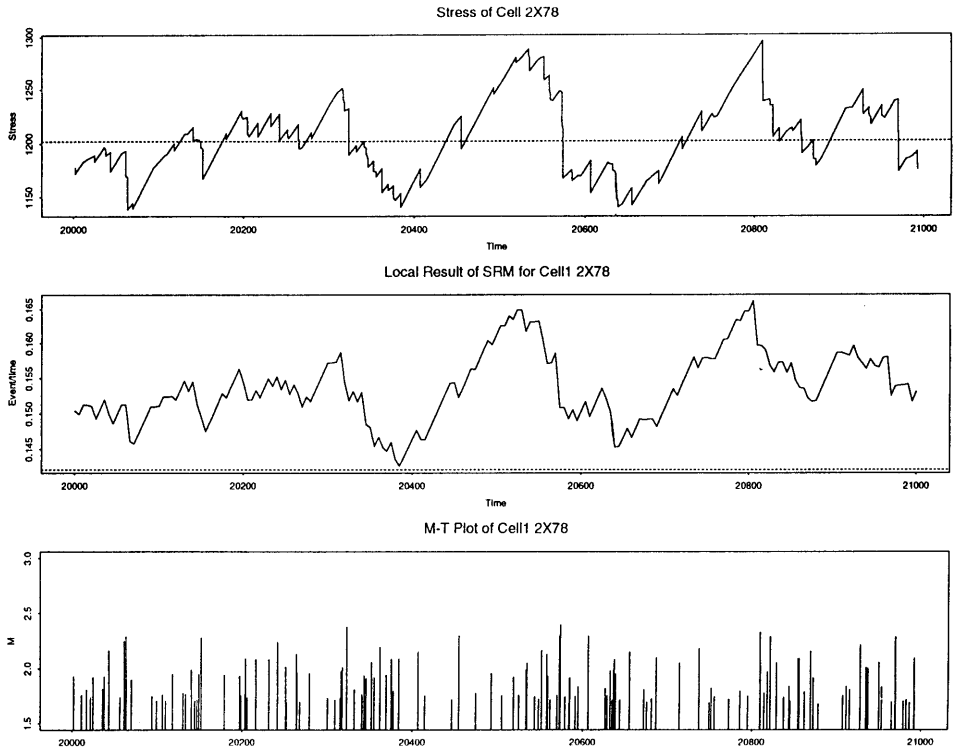

Figure 5
Computational results by using 141 events from relative time 20000 to 21000 in cellular automata model. (a) System stress in cellular automata model; (b) conditional probability intensity of earthquake occurrence calculated from stress release model (logarithm of the intensity is proportional to system stress); (c) M-t plot.

parameters, but, successfully simulates the stress variation of the system. AIC value of stress release model is smaller than that of Poisson model by 3.6, therefore, stress release model is superior to Poisson model for the synthetic catalog.

In practice, small earthquakes usually cannot be completely recorded. In order to study the effect of a catalog absent small events, we cut small events ($M < 2.2$) from 2,843 events during time 10,000 to 30,000 in our synthetic catalog, and keep only 262 relatively large events. Although this number is 10% of the total events, stress release model still can produce a satisfactory fit as shown in Figure 6. In this case, AIC value of stress release model is smaller than Poisson model by 7.4, evidently superior to Poisson model.

In actuality, we generally study a local region, instead of the entire system. In order to study the effect of the division of subregions, we cut the region parallel to the fault, and each subregion contains a complete fault. The results are shown in Figure 7. The stress release model now behaves even worse than Poisson model in

each subregion. Coupled stress release model has AIC value lower than Poisson model by 222.6, and lower than independent stress release model by 203.9. It still provides the best fit. The results indicate that coupled stress release model can significantly improve the fitting, although the stress release model may not work well in each subregion.

5. *Conclusion and Discussion*

Poisson model is often used in seismic hazard assessment. Comparing Poisson model with stress release model, the former is a common model with no physical meaning in statistics, but, the latter is a statistical model based on the geophysical model. The papers by ZHENG *et al.* (1991, 1994) have shown clearly that stress release model is obviously better than Poisson model. Based on these works, we

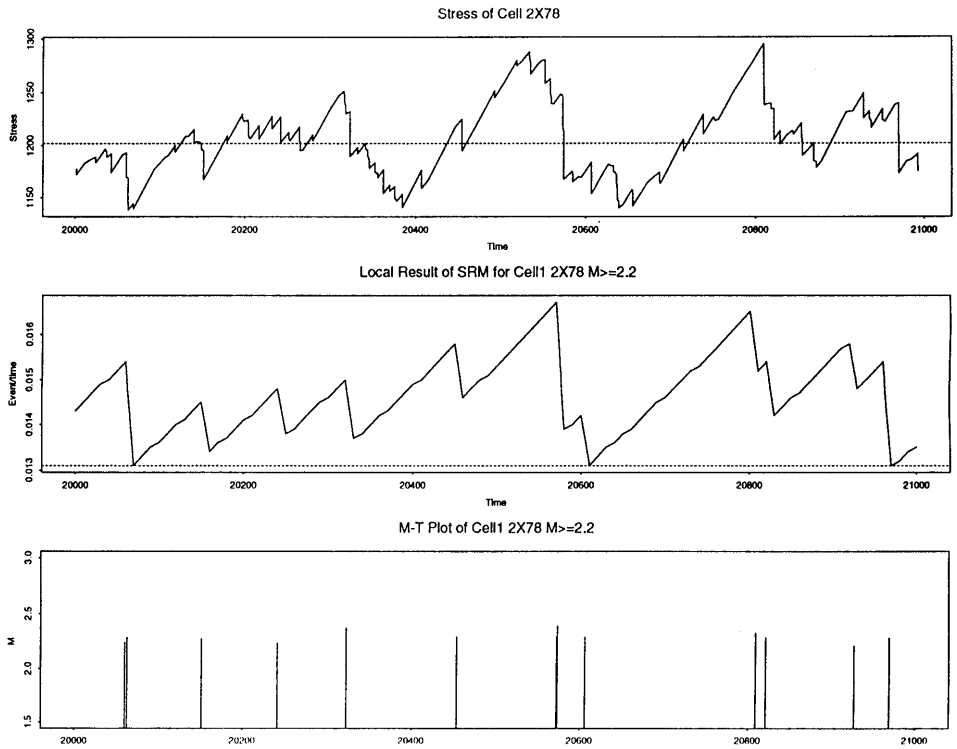

Figure 6
Computational results from 248 large earthquakes ($M > 2.2$) among 2,846 earthquakes in synthetic catalog from time 10000 to 30000. Only the curves within time period 20000–21000 are shown for comparison with Figure 5. (a) System stress in the cellular automata model; (b) conditional probability intensity of earthquake occurrence calculated from the stress release model; (c) M-t plot.

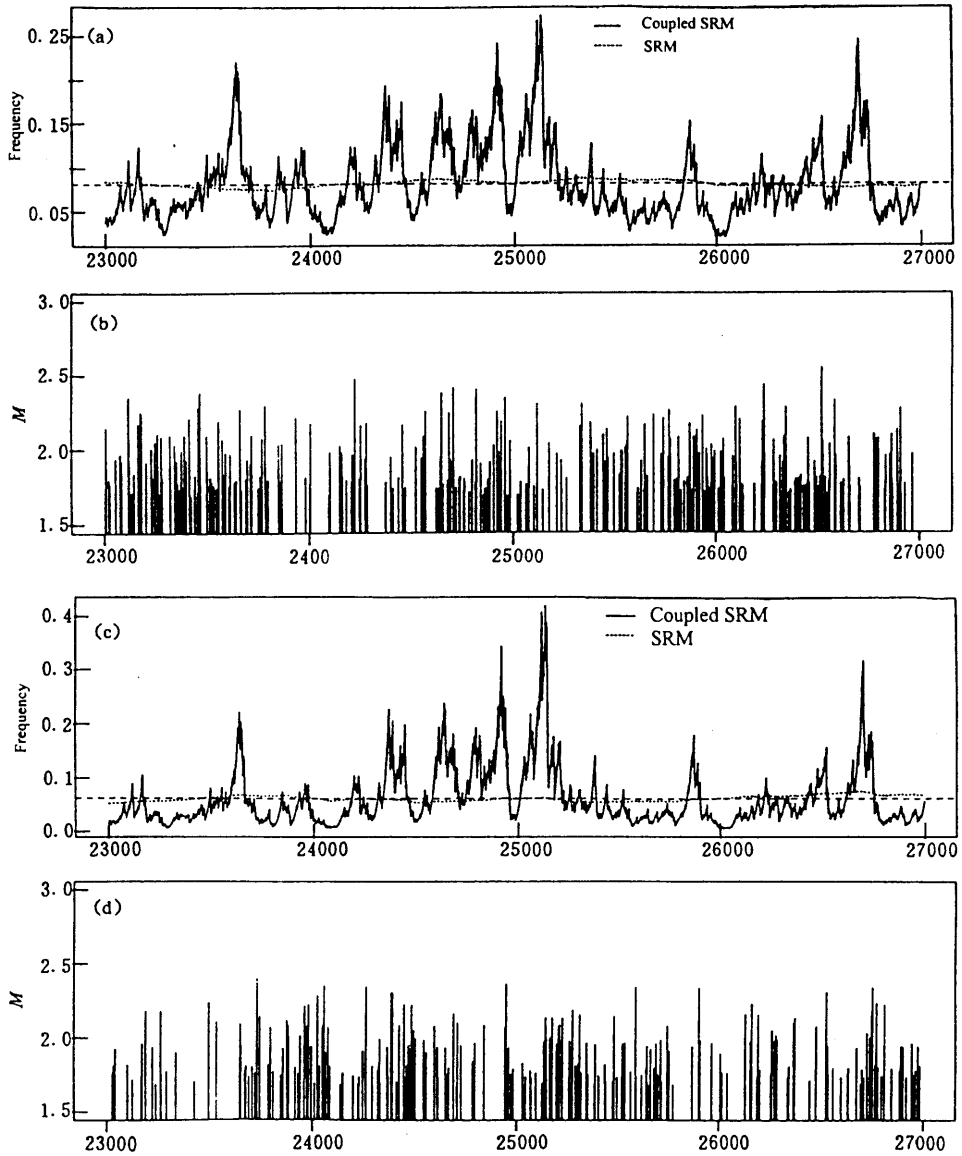

Figure 7

The results for a model with two subregions divided by a line parallel to the faults. (a) Conditional probability intensity of earthquake occurrence in the first subregion, calculated from coupled stress release model (solid line) and simple stress release model (dash line) respectively; (b) M-t plot for the first subregion; (c) conditional probability intensity of earthquake occurrence in the second subregion, calculated from coupled stress release model (solid line) and simple stress release model (dash line), respectively; (d) M-t plot for the second subregion.

have considered the characteristics of practical seismicity (interaction between different seismic subregions) and we have developed a coupled stress release model. The following research conclusions can be obtained:

(1) The interaction between different parts is an important factor when applying stress release model, i.e., coupled stress release model is better than common stress release model.

(2) The regional division according to seismicity characteristics is also a considerable problem when applying stress release model. The computation of stress release model in subregions is better than in the whole region.

(3) The regional stress can be simulated very well by our stress release model, which suggests that the model is a very useful tool in seismic hazard assessment.

Because stress release model only considers inner stress accumulation and inner stress release caused by earthquakes in the region, the values b and c of conditional probability intensity function λ in Equation (7) are always larger than 0. From a physical view, it means that the regional stress increases linearly with time and decreases when an earthquake occurs. Under the same condition, the values of b_1, b_2, c_{11} and c_{22} in coupled stress release model are also larger than 0, however the interaction between subregions can either decrease or increase the subregional stress, consequently, the values of c_{21} and c_{12} may be larger or less than 0. The positive values mean that the earthquakes in a subregion will cause stress release in the other subregion, and the negative values will cause stress increase. Because stress release model is a statistical model, we can determine the whole effect of interaction between two subregions using the results of coupled stress release model. In addition, if assuming $c_{12} = c_{21}$, its physical meaning is that the interaction way in two subregions and the corresponding time length of stress release (or increase) level caused by earthquakes in another subregion are identical. Although there is no definite conclusion between symmetrical CSRM and common CSRM, the symmetrical CSRM is not an adverse model.

Normalized magnitude m_0 is used only for simplifying computation, which is without physical meaning. It is unnecessary to use this term, but it will cause the values of stress drop S to become very large (value S is also just a comparative measure). In addition, when m_0 decreases one magnitude, the value c will increase $10^{0.75}$ times in Equations (7) and (10). The m_0 selection is not relative to the other parameters.

There are many similar assumptions in the stress release model and the cellular automata model of this paper. Both models assume stress linear increase of stresses during earthquake preparation, although the rates are different for each element in the cellular automata model and the entire system is assumed to have the same rate in the stress release model. The cellular automata has definite rules for stress adjustment during earthquake occurrence, and the stress release model also assumes every earthquake exerts a definite influence on system stress depending only on the magnitude of the earthquake. This similarity may explain why the stress release

model can provide a very good fit of system stress of the cellular automata model. Although the actual data is surely more complicated than the synthetic data, it is indicated by the fitting of the synthetic data of cellular automata that some system characteristics (the system stress in this study) can be predicted with a limited time, based on the catalog (or an incomplete catalog), even if the details of model and its parameters are not known.

When computing the accumulated stress release $S_j(t)$ in coupled stress release model, we think that the $S_j(t)$ of $\Sigma_{i=1}^{N_1(T)} \log \lambda_1(t_i)$ in Equation (11) (same as for $\Sigma_{j=1}^{N_2(T)} \log \lambda_2(t_j)$) just considers event time in the first subregion, which means that $S_j(t)$ is calculated up to time t_{i-1} for the ith event in the first subregion. The reason is that the log-likelihood method is to maintain balance between stress accumulation $b(t_i - t_{i-1})$ and stress release $\Sigma_{k=1}^{L_i} 10^{0.75(m_k - m_0)} + 10^{0.75(m_i - m_0)}$ during the time interval $[t_{i-1}, t_i]$, where L_i and m_k are separately the total number and magnitude of earthquakes of the second subregion which occurred in time interval $[t_{i-1}, t_i]$, m_i is the earthquake magnitude of the first subregion which occurred in time t_i.

Furthermore, we have also found problems in our work. These are of value for further works:

(1) The results obtained by coupled stress release models clearly raise the seismic occurrence probability of some main earthquakes (or a start earthquake in a cluster of earthquakes) from the conditional probability intensity plot, compared with the results obtained from independent stress release model and simple stress release model. It is very useful to predict these earthquakes using conditional probability intensity. However, on the other hand, the prediction for following earthquakes (not aftershocks) in a cluster of earthquakes by coupled stress release model is not good. In fact, the problem exists in all involved stress release models in the paper. One solution to this problem is to further improve the conditional probability intensity function in stress release model and to consider clustering earthquake activity in the function. The second method is to further divide the region and to allocate clustering earthquakes into each subregion.

(2) The regional division is very important in the stress release model. The region is divided into an area as small as possible, considering the condition of the tectonic environment. However, in practicality, because the reoccurrence time is prolonged, there will be fewer earthquakes in some subregions if they are too small, and fewer events will not satisfy the condition of the statistical calculation. For example, although the AIC value in common CSRM (or symmetrical CSRM) is less than that of other stress release models when the North China region is divided into four seismic belts (four subregions), the result is not in accord with the practical stress accumulation condition from the conditional probability intensity plot. The reason is that the value b is very large, causing stress increase rate to be excessively large (this is another reason that $b_1 = b_2$ CSRM is used in the paper). Thus, it is reasonable to decrease the lower limit of magnitude in order to increase the sample number in future research. Another method is to study stress release model using the synthetic earthquake catalog produced by the physical model.

(3) The stress release model does provide a means to estimate the temporal variation of system stress. It works very well for the synthetic catalog produced by cellular automata model. It is an interesting problem if this model also works very well for real earthquake data. Independent observations on regional stress changes are necessary.

(4) How to understand accumulated stress release $S(t)$ in coupled stress release model is worthy of further endeavor.

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Appendix

North China earthquake data set (1480–1996)

No.	Time	Lat.	Long.	Mag.	Reg.	No.	Time	Lat.	Long.	Mag.	Reg.
1	1484.01.29	40.4	116.1	6.75	E2	34	1704.09.28	34.9	106.8	6.00	W2
2	1487.08.10	34.3	108.9	6.25	W1	35	1709.10.14	37.4	105.3	7.50	W2
3	1501.01.19	34.8	110.1	7.00	W1	36	1718.06.29	35.0	105.2	7.50	W2
4	1502.10.17	35.7	115.3	6.50	E2	37	1720.07.12	40.4	115.5	6.75	E2
5	1536.10.22	39.6	116.8	6.00	E2	38	1730.09.30	40.0	116.2	6.50	E2
6	1548.09.13	38.0	121.0	7.00	E1	39	1739.01.03	38.8	106.5	8.00	W2
7	1556.01.23	34.5	109.7	8.00	W1	40	1815.10.23	34.8	111.2	6.75	W1
8	1561.07.25	37.5	106.2	7.25	W2	41	1820.08.03	34.1	113.9	6.00	E2
9	1568.04.25	39.0	119.0	6.00	E1	42	1829.11.19	36.6	118.5	6.00	E1
10	1568.05.15	34.4	109.0	6.75	W1	43	1830.06.12	36.4	114.2	7.50	E2
11	1573.01.10	34.4	104.1	6.75	W2	44	1831.09.28	32.8	116.8	6.25	E1
12	1587.04.10	35.2	113.8	6.00	E2	45	1852.05.26	37.5	105.2	6.00	W2
13	1597.10.06	38.5	120.0	7.00	E1	46	1861.07.19	39.7	121.7	6.00	E1
14	1604.10.25	34.2	105.0	6.00	W2	47	1879.07.01	33.2	104.7	8.00	W2
15	1614.10.23	37.2	112.5	6.50	W1	48	1882.12.02	38.1	115.5	6.00	E2
16	1618.05.20	37.0	111.9	6.50	W1	49	1885.01.14	34.5	105.7	6.00	W2
17	1618.11.16	39.8	114.5	6.50	E2	50	1888.06.13	38.5	119.0	7.50	E1
18	1622.03.18	35.5	116.0	6.00	E2	51	1888.11.02	37.1	104.2	6.25	W2
19	1622.10.25	36.5	106.3	7.00	W2	52	1920.12.06	36.7	104.9	8.50	W2
20	1624.02.10	32.4	119.5	6.00	E1	53	1922.09.29	39.2	120.5	6.50	E1
21	1624.04.17	39.8	118.8	6.25	E2	54	1937.08.01	35.4	115.1	7.00	E2
22	1624.07.04	35.4	105.9	6.00	W2	55	1945.09.23	39.5	119.0	6.25	E2
23	1626.06.28	39.4	114.2	7.00	E2	56	1966.03.22	37.5	115.1	7.20	E2
24	1627.02.15	37.5	105.5	6.00	W2	57	1967.03.27	38.5	116.5	6.30	E2

Appendix (Continued)

No.	Time	Lat.	Long.	Mag.	Reg.	No.	Time	Lat.	Long.	Mag.	Reg.
25	1634.01.15	34.1	105.3	6.00	W2	58	1969.07.18	38.2	119.4	7.40	E1
26	1642.06.30	35.1	111.1	6.00	W1	59	1975.02.04	40.7	122.8	7.30	E1
27	1654.07.21	34.3	105.5	8.00	W2	60	1976.04.06	40.2	121.1	6.20	W2
28	1658.02.03	39.4	115.7	6.00	E2	61	1976.07.28	39.4	118.0	7.82	E2
29	1665.04.16	39.9	116.6	6.50	E2	62	1976.09.23	39.9	106.4	6.20	W1
30	1668.07.25	35.3	118.6	8.60	E1	62	1979.08.25	41.2	108.1	6.00	W2
31	1679.09.02	40.0	117.0	8.00	E2	64	1989.10.18	40.0	113.7	6.00	W1
32	1683.11.22	38.7	112.7	7.00	W1	65	1996.05.03	40.8	109.6	6.50	W2
33	1695.05.18	36.0	111.5	8.00	W1						

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